Dohyun **Kim**

FINITE ELEMENT METHOD · PDE-CONSTRAINED OPTIMIZATION · HIGH-PERFORMANCE COMPUTING

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Summary

Dohyun Kim is an applied mathematician with a strong background in functional analysis, numerical analysis, and scientific computing. His main research interests include, but not limited to, PDE-constrained optimization, Riemannian optimization, and high-performance computing. His research goal is to develop efficient and scalable discrete algorithms for solving large-scale optimization problems arising from various scientific and engineering applications, such as optimal control with operational/safety constraints.

Education

Yonsei University — *Seoul, S.Korea*

PH.D. IN COMPUTATIONAL SCIENCE AND ENGINEERING — *Mar. 2015 - Feb. 2021*

- Thesis: "Non-conforming methods for 4th-order PDEs"
- Advisor: Prof. Eun-Jae Park
- Polytopal finite element methods, discontinuous Galerkin methods, a priori/a posteriori error analysis

Hanyang University — *Seoul, S.Korea*

B.S. IN MATHEMATICS — *Mar. 2011 - Feb. 2015*

- Basis of functional analysis, linear algebra, numerical analysis, and graph theory.

Professonal Experience

Brown University — *Providence, RI, USA*

RESEARCH ASSOCIATE, POSTDOC — *Jan. 2023 - Present*

- PDE-constrained optimization
- A priori/a posteriori error analysis for variational inequaltiies

Yonsei University — *Seoul, S.Korea*

POSTDOC — *Sep. 2022 - Dec. 2022*

- Polytopal finite element methods

Hong Kong Centre for Cerebro-cardiovascular Health Engineering (COCHE) — *Hong Kong, China*

POSTDOC — *Oct. 2021 - Aug. 2022*

- Pressure robust numerical methods for Navier-Stokes equations
- Fluid-structure interaction problems

Yonsei University — *Seoul, S.Korea*

POSTDOC — *Mar. 2021 - Oct. 2021*

- Polytopal finite element methods for Stokes equations

Teaching Experience

Numerical Solution of Partial Differential Equations II
Co-instructor - APMA2560

Brown University — *Spring 2026*

- Recitation instructor (theory and C++ implementation of finite element methods)

Introduction to Computational Linear Algebra
Instructor - APMA1170

Brown University — *Fall 2024*

- Linear algebra (vector spaces, linear transformations, eigenvalues)
- Numerical linear algebra (LU decomposition, Krylov subspace methods, eigenvalue solvers)

Numerical Solution of Partial Differential Equations II
Teaching Assistant - APMA2560

Brown University — *Spring 2024*

- Recitation instructor (theory and C++ implementation of finite element methods)

Publications

Preprints

[1] **The SiMPL Method for Multi-Material Topology Optimization** — *2026*
P. Gangl, B. Keith, **D. Kim**, and T. M. Surowiec
arXiv:2605.11994

[2] **Proximal discontinuous Galerkin methods for variational inequalities** — *2026*
A. Ern, B. Keith, **D. Kim**, R. Masri, and B. Riviere
arXiv:2604.19708

[3] **The proximal Galerkin method for non-symmetric variational inequalities** — *2026*
G. Fu, B. Keith, **D. Kim**, R. Masri, and W. Pazner
arXiv:2602.14967

[4] **Pressure-robust staggered DG methods for the Navier-Stokes equations on general meshes** — *2021*
D. Kim, L. Zhao, E. Chung, and E.-J. Park
arXiv:2107.09226

Journal Articles

[1] **Finite element approximation to the non-stationary quasi-geostrophic equation** — *2025*
D. Kim, A. K. Pani, and E.-J. Park
ESAIM: Mathematical Modelling and Numerical Analysis, 59(4):1973–2003

[2] **Analysis of the SiMPL method for density-based topology optimization** — *2025*
B. Keith, **D. Kim**, B. S. Lazarov, and T. M. Surowiec
SIAM Journal on Optimization, 35(2):1134–1164

[3] **A simple introduction to the SiMPL method for density-based topology optimization** — *2025*
D. Kim, B. S. Lazarov, T. M. Surowiec, and B. Keith
Structural and Multidisciplinary Optimization, 68(4):1–17

[4] **High-performance finite elements with MFEM** — *2024*
J. Andrej, N. Atallah, J.-P. Bäcker, J.-S. Camier, D. Copeland, V. Dobrev, Y. Dudouit, T. Duswald, B. Keith, **D. Kim**, T. Kolev, B. Lazarov, K. Mittal, W. Pazner, S. Petrides, S. Shiraiwa, M. Stowell, and V. Tomov
The International Journal of High Performance Computing Applications, 38(5)

[5] **DynAMO: Multi-agent reinforcement learning for dynamic anticipatory mesh optimization with applications to hyperbolic conservation laws** *2024*

T. Dzanic, K. Mittal, **D. Kim**, J. Yang, S. Petrides, B. Keith, and R. Anderson
Journal of Computational Physics, 506:112924

[6] **Staggered DG method with small edges for Darcy flows in fractured porous media** *2022*

L. Zhao, **D. Kim**, E.-J. Park, and E. Chung
Journal of Scientific Computing, 90(3):83

[7] **Morley finite element methods for the stationary quasi-geostrophic equation** *2021*

D. Kim, A. K. Pani, and E.-J. Park
Computer Methods in Applied Mechanics and Engineering, 375:113639

[8] **Review and implementation of staggered DG methods on polygonal meshes** *2021*

D. Kim, L. Zhao, and E.-J. Park
Journal of the Korean Society for Industrial and Applied Mathematics, 25(3)

[9] **Staggered DG methods for the pseudostress-velocity formulation of the Stokes equations on general meshes** *2020*

D. Kim, L. Zhao, and E.-J. Park
SIAM Journal on Scientific Computing, 42(4):A2537–A2560

[10] **Error estimates of B-spline based finite-element methods for the stationary quasi-geostrophic equations of the ocean** *2018*

D. Kim, T. Y. Kim, E.-J. Park, and D.-w. Shin
Computer Methods in Applied Mechanics and Engineering, 335:255–272

Open source contributions

[1] **General hyperbolic class and Refactored Ex18, Ex18p** *2024*

MFEM (C++ library for FEM managed by LLNL)
Flux evaluation for general hyperbolic PDEs and refactored examples for the compressible Euler equations

[2] **Refactoring Ex18 with preassembly option** *2024*

MFEM (C++ library for FEM managed by LLNL)
Option for preassembly of the nonlinear operator for optimized performance

[3] **Enhancement of AutoDiff miniapp** *2025 - WIP*

MFEM (C++ library for FEM managed by LLNL)
Added support for automatic differentiation in nonlinear problems and miniapp examples

[4] **Add `SetEssentialTDofs` to `(Par)BlockNonlinearForm`** *2025*

MFEM (C++ library for FEM managed by LLNL)
Added support for essential boundary conditions in nonlinear block systems

Presentation

Conference organized

Mini-symposium: Recent developments in mathematical analysis and numerics for incompressible flow and related problems *Pittsburgh, Pennsylvania, USA*
2022 SIAM Annual Meeting *July, 2022*

Special Session on Constrained Optimization and Optimal Control *Lawrence, Kansas, USA*
2025 Spring Central Sectional Meeting *April, 2025*

Selected talks (among 20+ talks)

A Projected Mirror Descent Method for Density-based Multi-material Topology Optimization *Atlanta, Georgia, USA*
INFORMS Optimization Society Conference *March, 2026*

SiMPL method for Topology Optimization *Fort Worth, Texas, USA*
SIAM Conference on Computational Science and Engineering *March, 2025*

SiMPL Method for Density Based Topology Optimization *Worcester, Massachusetts, USA*
Worcester Polytechnic Institute *October, 2024*

Dynamic anticipatory mesh optimization with reinforcement learning *Larnaca, Cyprus*
International Conference on High Order Finite Element and Isogeometric Methods *May, 2023*

Pressure-Robust Staggered DG Method for Navier-Stokes Equations *Virtual*
SIAM Conference on Analysis of Partial Differential Equations *March, 2022*

Peer Review & Research Visit

Reviewer, Computer & Mathematics with Applications
Reviewer, SIAM Journal on Scientific Computing
Research Visit, Simula Research Laboratory: Topology Optimization *Oslo, Norway* *Dec. 4–8, 2023*
Research Visit, Technische Universität München: Hyperbolic Conservation Laws on Manifold *Munich, Germany* *June 10–14, 2024*